

# JING LIU

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School of Physical Science and Technology, Beijing University of Posts and Telecommunications  
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## EDUCATION

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School of Systems Science, Beijing Normal University 2018.09 – 2023.07  
*Ph.D.*, supervisor: [Jiang Zhang](#)

Department of Physics, Beijing Normal University 2014.09 – 2018.07  
*Bachelor of Science*

## WORKING EXPERIENCE

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School of Physical Science and Technology, BUPT 2025.11 – Present  
*Tenure-Track Assistant Professor*

Institute of Theoretical Physics, Chinese Academy of Sciences 2023.11 – 2025.10  
*Postdoctoral Researcher*, postdoc collaborator: [Pan Zhang](#)

Zhejiang Lab 2023.07 – 2023.10  
*Senior Research Specialist*

## RESEARCH GRANTS

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Young Scientists Fund of the NSFC 2025.01 - 2027.12

## PUBLICATIONS

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- Universal Sampling of Spin Systems across Quenched Disorder  
**Jing Liu**, Yeyuan Wu, Ying Tang, Pan Zhang  
[arXiv:2609.11336](#)
- Tensor-Network Markov-Chain Monte Carlo: Efficient Sampling of Three-Dimensional Spin Glasses  
Tao Chen, **Jing Liu**<sup>†</sup>, Youjin Deng<sup>†</sup>, Pan Zhang<sup>†</sup> (<sup>†</sup>corresponding author)  
[Physical Review Letters 137 \(11\), 117101](#)
- Variational Monte Carlo with Row-Update Projected Entangled-Pair States and Its Applications to Quantum Spin Glasses  
Tao Chen, **Jing Liu**, Yantao Wu, Pan Zhang, Youjin Deng  
[Physical Review B 114 \(14\), 144201](#)
- Characterizing Full Nonequilibrium Dynamics of Simple Exclusion Processes  
Zhima Liu, **Jing Liu**, Pan Zhang, Ying Tang  
[arXiv:2608.25606](#)
- Exact Autoregressive Sampling of Planar Ising Spin Glasses via the Kac–Ward Theory  
**Jing Liu**, Tao Chen, Tianrui Che, Lei Wang, Youjin Deng, Pan Zhang  
[arXiv:2608.24382](#)
- Scalable Physics-Inspired Transformers for Spin Glasses  
Lu Zhong, Wenli Duan, **Jing Liu**<sup>†</sup>, Pan Zhang<sup>†</sup>, Ying Tang<sup>†</sup> (<sup>†</sup>corresponding author)  
[arXiv:2606.22984](#)

7. Tracking Large Chemical Reaction Networks and Rare Events by Neural Networks  
Jiayu Weng, Xinyi Zhu, **Jing Liu**, Linyuan Lü, Pan Zhang, Ying Tang  
[arXiv:2512.10309](#)
8. BatchTNMC: Efficient sampling of two-dimensional spin glasses using tensor network Monte Carlo  
Tao Chen, Jingtong Zhang, **Jing Liu**<sup>†</sup>, Youjin Deng<sup>†</sup>, Pan Zhang<sup>†</sup> (<sup>†</sup>corresponding author)  
[arXiv:2509.19006](#)
9. Efficient optimization of variational autoregressive networks with natural gradient  
**Jing Liu**, Ying Tang, Pan Zhang  
[Physical Review E 111 \(2\), 025304](#)
10. Recurrent Dynamic Message Passing with Loops for Epidemics on Networks  
Fei Gao, **Jing Liu**, Yaqian Zhao  
[Chaos, Solitons & Fractals 184, 115045](#)
11. Learning nonequilibrium statistical mechanics and dynamical phase transitions  
Ying Tang\*, **Jing Liu**\*, Jiang Zhang, Pan Zhang (\*contributed equally)  
[Nature Communications 15 \(1\), 1117](#)
12. Tensor networks for unsupervised machine learning  
**Jing Liu**, Sujie Li, Jiang Zhang, Pan Zhang  
[Physical Review E 107 \(1\), L012103](#)
13. Gumbel-Softmax-Based Optimization: A Simple General Framework for Optimization Problems on Graphs  
Yaixin Li, **Jing Liu**, Guozheng Lin, Yueyuan Hou, Muyun Mou, Jiang Zhang  
[Computational Social Networks 8 \(1\), 5](#)
14. Human mobility in interest space and interactive random walk  
Fanqi Zeng, Li Gong, **Jing Liu**, Jiang Zhang, Qinghua Chen, Ruyue Xin  
[Journal of Physics: Complexity 1 \(2\), 025004](#)
15. Gumbel-Softmax Optimization: A Simple General Framework for Combinatorial Optimization Problems on Graphs  
**Jing Liu**, Fei Gao, Jiang Zhang  
[COMPLEX NETWORKS 2019: Complex Networks and Their Applications VIII pp 879–890](#)
16. A general deep learning framework for network reconstruction and dynamics learning  
Zhang Zhang, Yi Zhao, **Jing Liu**, Shuo Wang, Ruyi Tao, Ruyue Xin, Jiang Zhang  
[Applied Network Science 4 \(110\), 1-17](#)
17. Chinese Address Similarity Calculation Based on Auto Geological Level Tagging  
**Jing Liu**, Jianbin Wang, Changqing Zhang, Xiubo Yang, Jianbo Deng, Ruihe Zhu, Xiaojie Nan, Qinghua Chen  
[ISNN 2019: Advances in Neural Networks – ISNN 2019 pp 431–438](#)
18. Coevolution of agent's behavior and noise parameters in majority vote game on multilayer networks  
**Jing Liu**, Ying Fan, Jiang Zhang, Zengru Di  
[New Journal of Physics 21 \(1\), 015007](#)